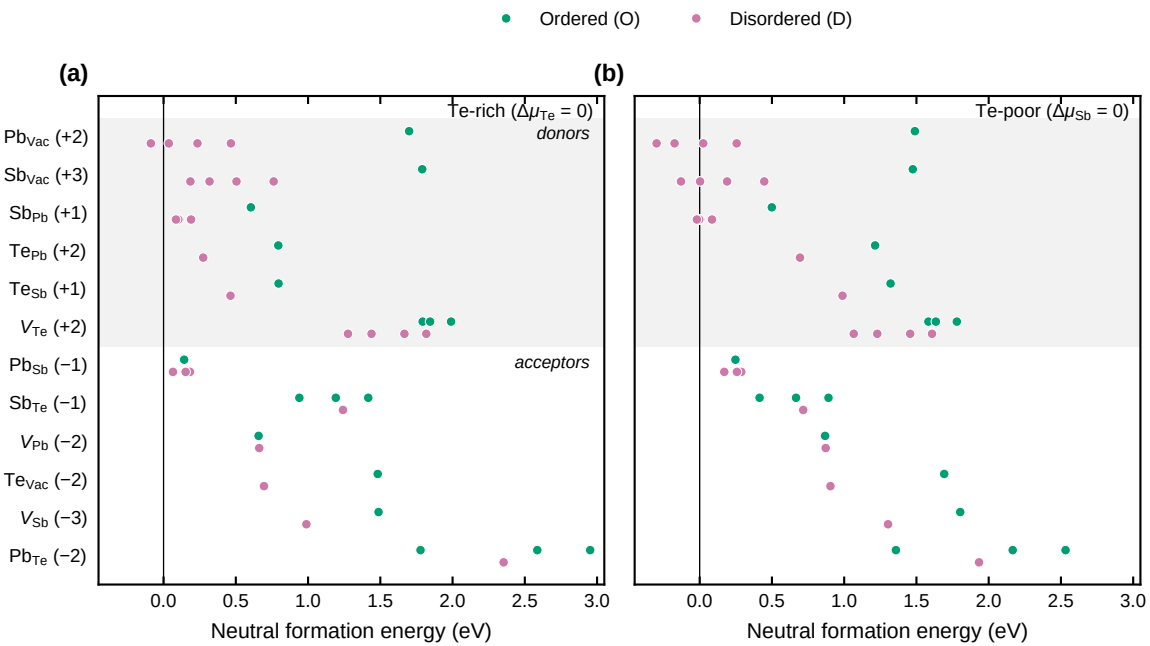
Ordered (O)
Disordered (D)
(a)
(b)
Te-rich ($\Delta\mu_{Te}$ = 0)
Te-poor ($\Delta\mu_{Sb}$ = 0)
donors
acceptors
Pb_{Vac} (+2)
Sb_{Vac} (+3)
Sb_{Pb} (+1)
Te_{Pb} (+2)
Te_{Sb} (+1)
V_{Te} (+2)
Pb_{Sb} (−1)
Sb_{Te} (−1)
V_{Pb} (−2)
Te_{Vac} (−2)
V_{Sb} (−3)
Pb_{Te} (−2)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Neutral formation energy (eV)